

Model Comparison Matrix

This document gives a single-place reference for the current model stack: what each model does, how it is used, and how it co

The metrics below are read from the saved bundle metadata in web/model-stats.json and the current model bundles.

1. Bundle Inventory

Bundle	Modality	Working purpose	Runtime role	Primary input	Current status
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text_transformer	Text	Transformer-based multilabel text scorer for narrative screening	Primary text scorer	Free text / tran	
text	Text	Classical text bundle for fallback screening	Fallback text scorer	Text features	Active
audio	Audio	Main acoustic screening bundle for mood-heavy signals	Primary audio scorer	Acoustic summary features	A
audio_spectrogram	Audio	CNN on spectrogram features	Secondary audio stabilizer	Spectrogram tensor	Active
audio_sequence	Audio	BiLSTM on chunk sequences	Fallback audio stabilizer	Sequence features	Active
image_dl	Image	CNN-based image scorer with stronger augmentation and diversity	Primary image scorer	Face/image te	
image	Image	Classical face-feature bundle	Fallback image scorer	Face-derived features	Active
comorbidity	Multimodal	Classifier-chain ensemble for joint label interaction modeling	Downstream comorbidity head	Text	

2. Current Comparison Matrix

Bundle	Sample count	Macro F1	Label accuracy	Exact match	Strengths	Weak spots
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text	420,000	0.5956	-	-	Large training footprint, stable fallback text scorer	Classical features only, less expressive than
text_transformer	8,000	0.5762	0.7621	0.3731	Better language modeling and better label-specific behavior on current te	
audio	10,080	0.6019	-	-	Best overall audio bundle, strongest practical audio scorer	Mood-heavy domains still need car
audio_spectrogram	1,082	0.5328	0.6715	0.2074	Helpful secondary acoustic view	Not yet better than main audio bund
audio_sequence	1,440	0.3893	0.4940	0.0069	Captures temporal chunk patterns	Weakest audio branch, fallback only
image	3,360	0.6019	-	-	Stable classical image fallback	Less expressive than the DL image bundle
image_dl	395	0.6168	0.7215	0.2152	Best current image scorer	Needs more data diversity to generalize further
comorbidity	24	0.4081	0.6786	0.3750	Joint label interaction model, now blended with upstream modality consensus	W

3. Runtime Priority Order

The runtime currently prefers:

1. text_transformer over text
2. audio over audio_spectrogram and audio_sequence
3. image_dl over image
4. comorbidity after upstream text/audio/image consensus is available

4. Practical Interpretation

- text_transformer is the best text choice when installed, but the classical text bundle remains a useful fallback and can still be

- audio is the main audio scorer, while the spectrogram and sequence branches are secondary stabilizers.
- image_dl is currently the strongest image bundle and should be treated as the default image scorer.
- comorbidity is intentionally last in the stack and should be retrained after text, audio, and image stabilize.

5. How To Use This Matrix

Use this file when you need to answer questions like:

- Which model is primary for a modality?
- Which model is only a fallback?
- Which bundle has the weakest current validation signal?
- Which model should be retrained first?

6. Related Documentation

- README.md (./README.md)
- docs/ARCHITECTURE.md (ARCHITECTURE.md)
- docs/MULTIMODAL_FUSION.md (MULTIMODAL_FUSION.md)
- docs/PROJECT_DOCUMENTATION.md (PROJECT_DOCUMENTATION.md)
- docs/DEEP_LEARNING_MODEL_GUIDE.md (DEEP_LEARNING_MODEL_GUIDE.md)